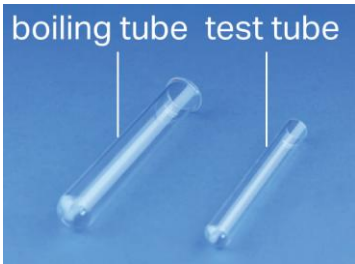
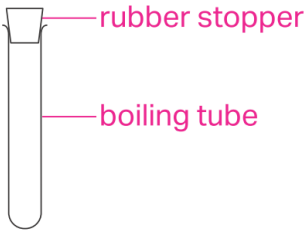
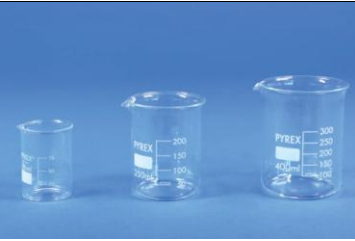
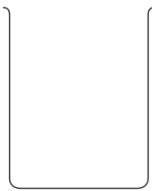
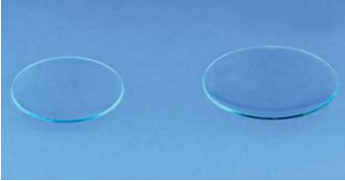
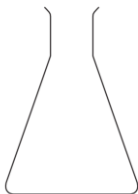
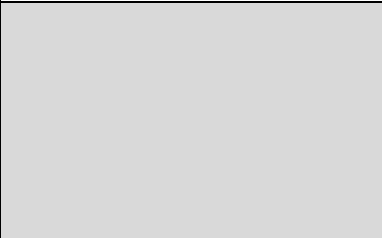
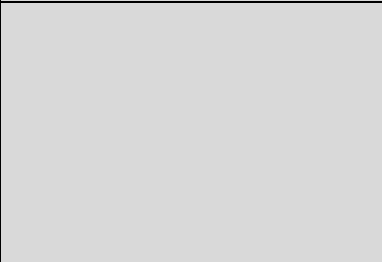


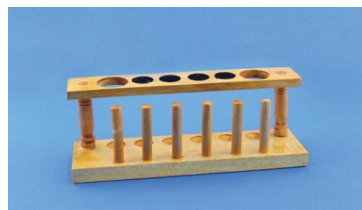
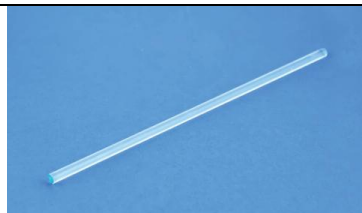
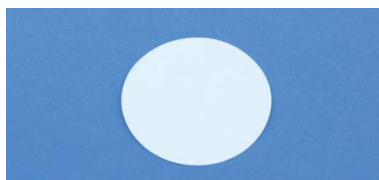
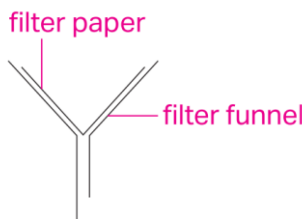
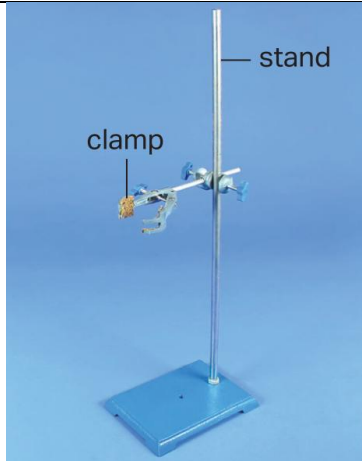
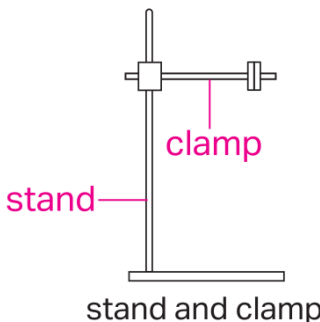
Chapter 1 – Introduction to Science

Apparatus & cross-sectional diagrams

Apparatus 儀器

Test tube 試管			To hold a smaller amount of liquids or solids (用於盛裝較少量的液體或固體)
Boiling tube 沸騰管			To hold a larger amount of liquids or solids (用於盛裝較大量的液體或固體)
Beaker 燒杯			To hold liquids or solids (用於盛裝液體或固體)
Measuring cylinder 量筒			With marks 刻度 to measure the volume of liquids (用於測量液體的體積)
Gas jar 集氣瓶			To collect or hold gases (用於收集或盛裝氣體)
Watch glass 錶玻璃			To hold a small amount of liquids or solids (用於盛裝少量液體或固體)

Evaporating dish 蒸發皿			To hold a liquid to be evaporated (用於盛裝需蒸發的液體)
Conical flask 錐形瓶			To hold liquids or solids, can be used in mixing solution (用於盛裝液體或固體，可以用於調和)
Reagent bottle 試劑瓶			To store liquids (用於儲存液體)
Dropping bottle 滴瓶			To store liquids that will be used in drops (用於儲存需以滴狀使用的液體)
Dropper 滴管			To transfer a small amount of liquids (用於轉移少量液體)
Spatula 刮勺			To transfer a small amount of powder (用於轉移少量粉末)
Tongs 鉗子			To hold hot objects (用於夾取熱物體)

Test tube holder 試管夾			To hold a test tube when heating (用於加熱時夾持試管)
Test tube rack 試管架			To hold test tubes or boiling tubes safely (用於安全地存放試管或沸騰管)
Glass rod 玻璃棒			To stir liquids (用於攪拌液體)
Filter paper 濾紙			To separate solids from liquids during filtration (用於過濾時分離固體與液體)
Filter funnel 漏斗			To let liquids flow down during filtration (用於過濾時讓液體流下)
Stand and clamp 支架和夾具			To fix the position of apparatus (用於固定儀器的位置)

A. Spelling exercises

Section 1: Basic Containers

1. Holds small liquids/solids for experiments: **T** _____ **T** _____
(用於盛裝少量液體/固體的細長玻璃管)
2. Thick-walled tube for boiling liquids: **B** _____ **T** _____
(用於煮沸液體的厚壁管)
3. Wide container for mixing solutions: **B** _____
(用於混合溶液的廣口容器)
4. Measures exact liquid volumes: **M** _____ **C** _____
(帶刻度測量液體體積的儀器)
5. Collects/stores gases: **G** _____ **J** _____
(收集/儲存氣體的專用瓶)

Section 2: Liquid Handling

6. Transfers liquid drops: **D** _____
(轉移液滴的塑膠工具)
7. Stores liquid reagents: **R** _____ **B** _____
(儲存化學試劑的棕色玻璃瓶)
8. Evaporates liquids: **E** _____ **D** _____
(蒸發液體的陶瓷器皿)
9. Holds small samples: **W** _____ **G** _____
(盛放少量樣品的弧形玻璃)
10. Mixes solutions in labs: **C** _____ **F** _____
(實驗室混合溶液的錐形瓶)

Section 3: Heating Tools

11. Grips hot objects: T_____

(夾取高溫物體的金屬工具)

12. Holds tubes during heating: T_____ T_____ H_____

(加熱時固定試管的夾子)

13. Directs liquid flow: F_____ F_____

(過濾時引導液體流動的漏斗)

Section 4: Other Tools

14. Holds multiple tubes: T_____ T_____ R_____

(垂直存放多支試管的架子)

15. Apparatus support frame: S_____ & C_____

(用於固定儀器的位置的支架組合)

16. Separates solids from liquids: F_____ P_____

(過濾時分離固液的多孔紙)

17. Transfers powders: S_____

(轉移粉末的不銹鋼刮勺)

18. Seals containers: R_____ S_____

(密封瓶口的橡膠部件)

19. Stores dropper liquids: D_____ B_____

(儲存滴加試劑的專用瓶)

20. Stirring tool: G_____ R_____

(攪拌溶液的玻璃棒)

B. MCQ

1. Which of the following apparatus can be used for holding about 2 cm^3 of water?

A. Test tube



B. Beaker



C. Filter funnel



2. Which of the following apparatus can be used for measuring 15 cm^3 of water?

A. Reagent bottle



B. Beaker



C. Measuring cylinder



3. Which of the following apparatus can be used for stirring liquids?

A. Spatula



B. Glass rod



C. Dropper



4. Which of the following apparatus can be used for transferring a small amount of liquid?

A. Test tube



B. Dropper



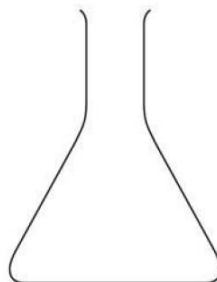
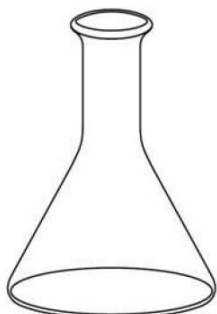
C. Watch glass



Drawing vertical section diagrams 剖面圖

Incorrect...

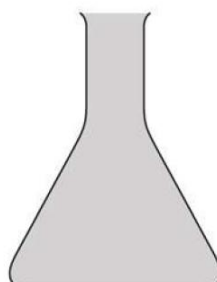
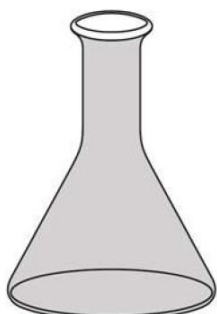
Should NOT be 3D



Correct!

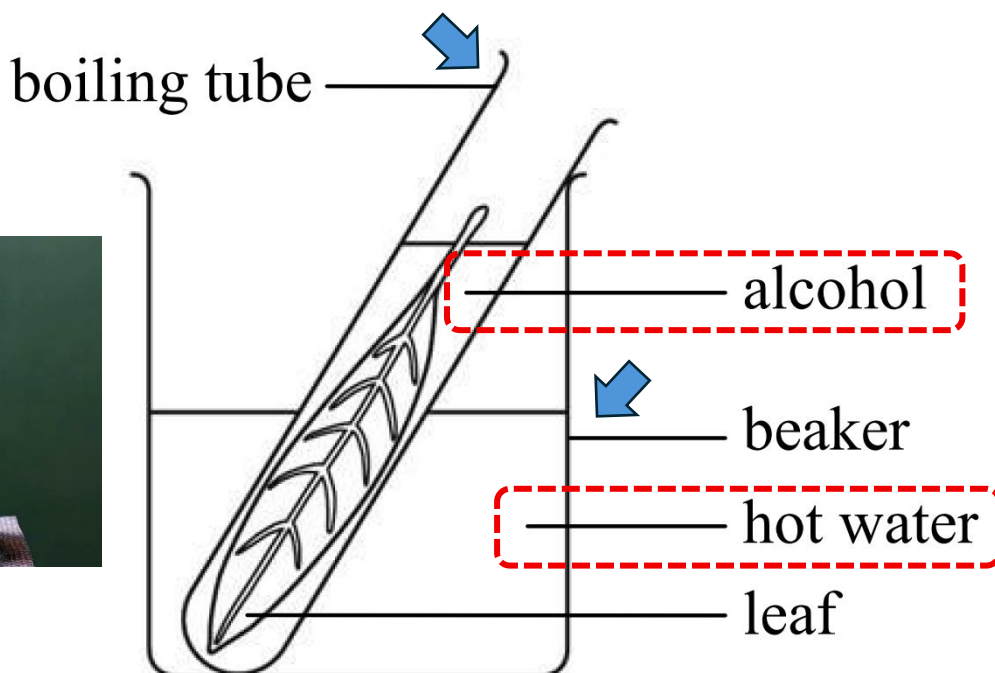
Incorrect...

Should NOT be 3D and should NOT shade



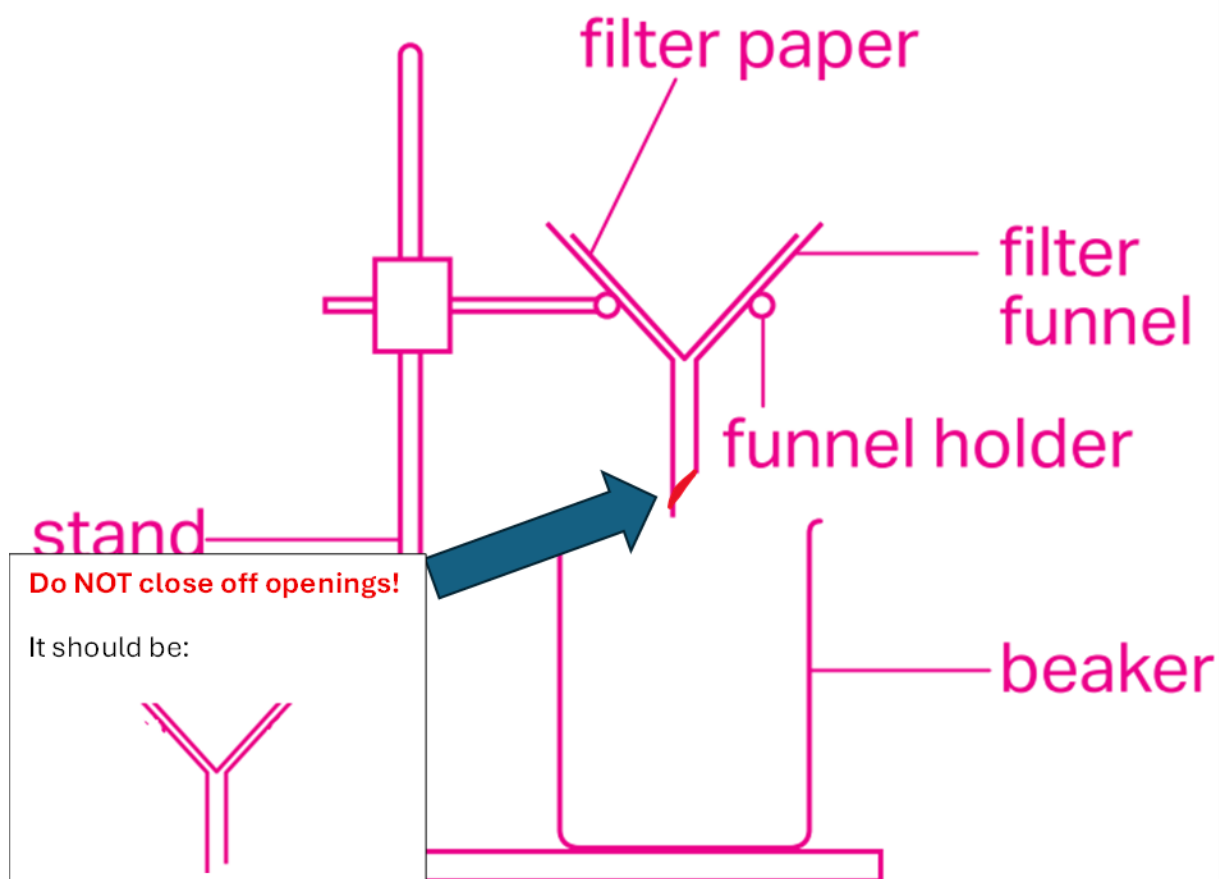
Incorrect...

Should NOT shade



- ◆ For labels of **containers**, straight lines should be **touching the wall**.
- ◆ For labels of **liquids**, straight lines should be **pointing inside**.

You should only use ONE LINE! Or it would be a double-walled / broken container



C. Drawing exercise

1. Draws the vertical section diagram of the bellowing setup

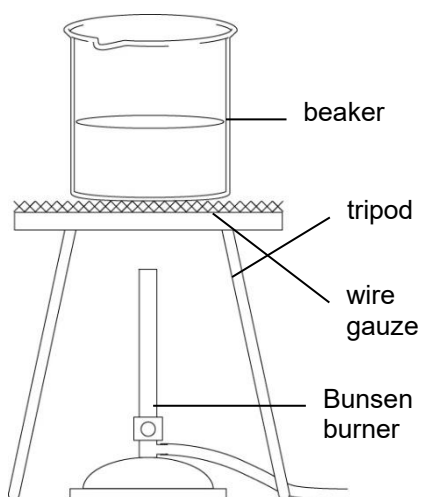


2. Mary draws a vertical section diagram for the experimental set-up as shown below. However, there are some mistakes. Circle the mistakes and draw the correct vertical section diagram in the space provided.

a)

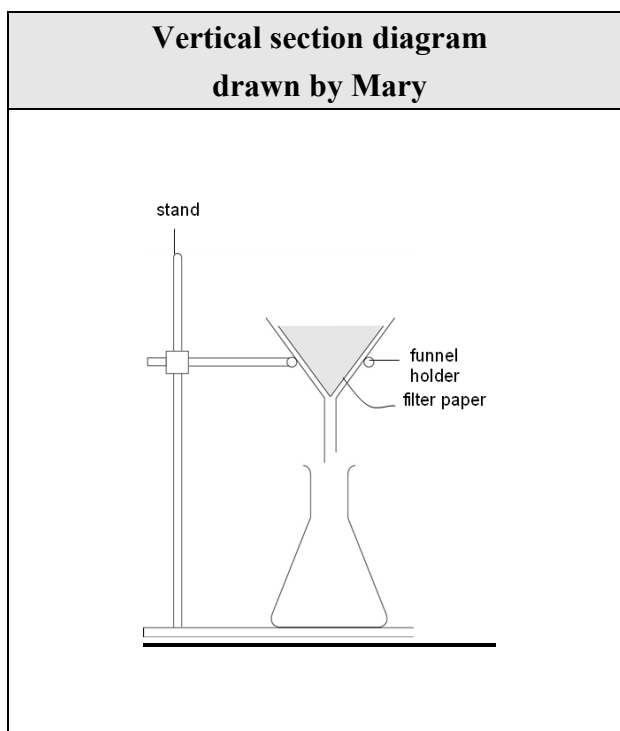
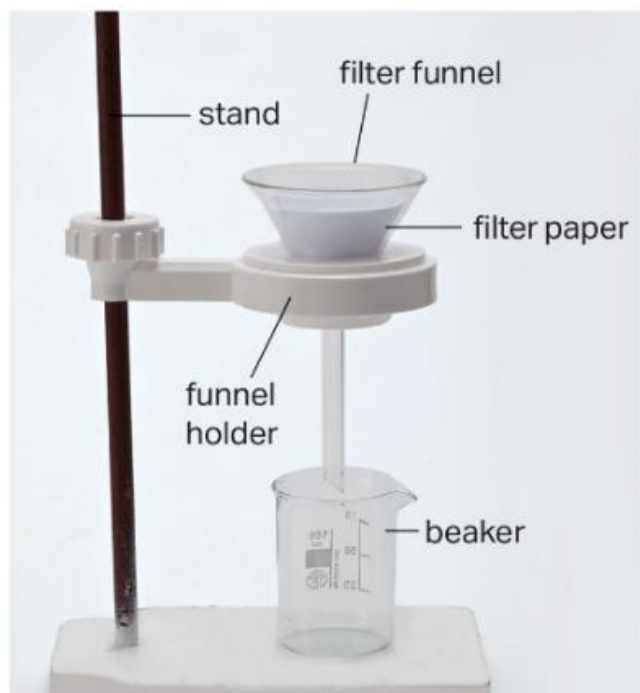


Vertical section diagram
drawn by Mary



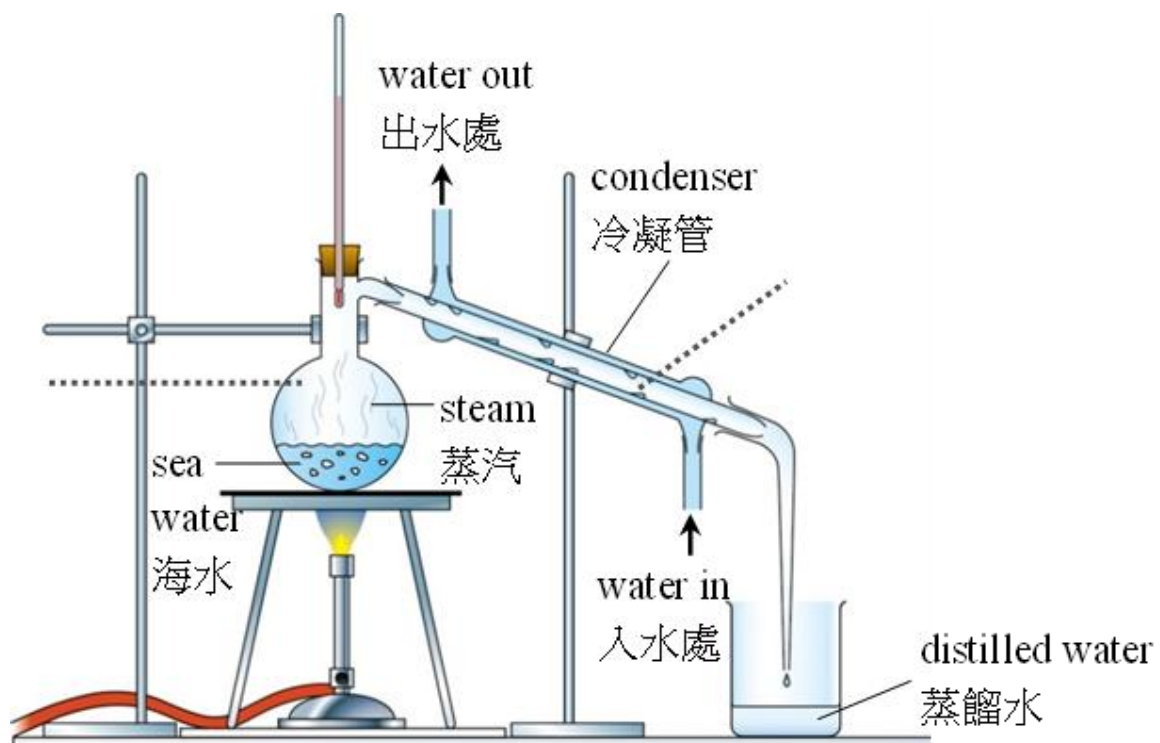
Correct vertical section diagram

b)



Correct vertical section diagram

3. Draws the vertical section diagram of the bellowing setup



Answer Key

Apparatus 儀器

A. Spelling exercises

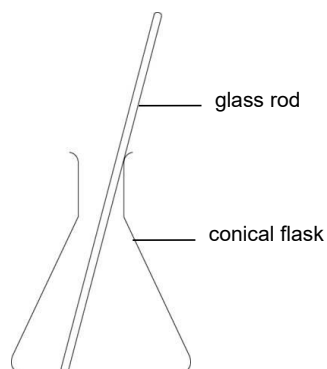
1. Test Tube (试管)
2. Boiling Tube (沸腾管)
3. Beaker (烧杯)
4. Measuring Cylinder (量筒)
5. Gas Jar (集气瓶)
6. Dropper (滴管)
7. Reagent Bottle (试剂瓶)
8. Evaporating Dish (蒸发皿)
9. Watch Glass (表面皿)
10. Conical Flask (锥形瓶)
11. Tongs (钳子)
12. Test Tube Holder (试管夹)
13. Filter Funnel (漏斗)
14. Test Tube Rack (试管架)
15. Stand & Clamp (支架和夹子)
16. Filter Paper (滤纸)
17. Spatula (刮勺)
18. Rubber Stopper (橡胶塞)
19. Dropping Bottle (滴瓶)
20. Glass Rod (玻璃棒)

B. MCQ

A C B B

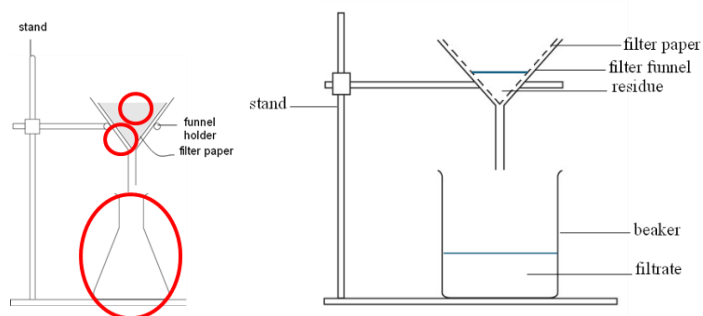
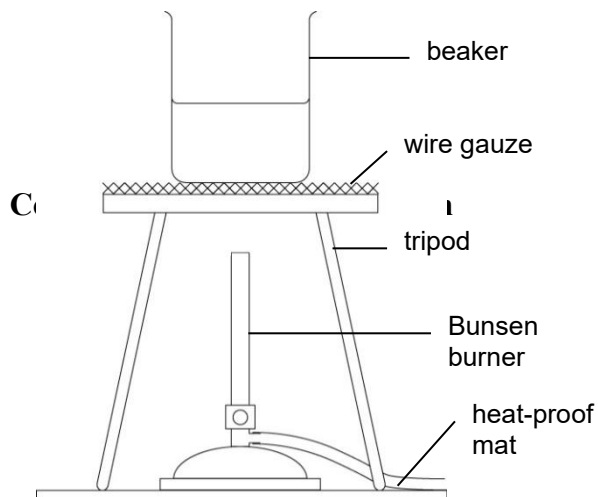
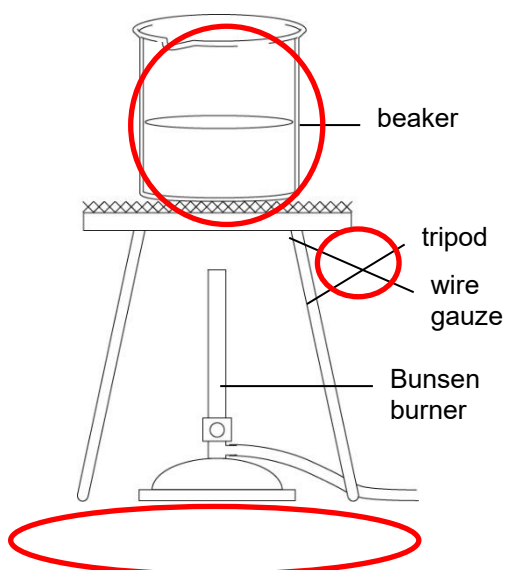
C. Drawing exercise

1.



2.

a)



3.

